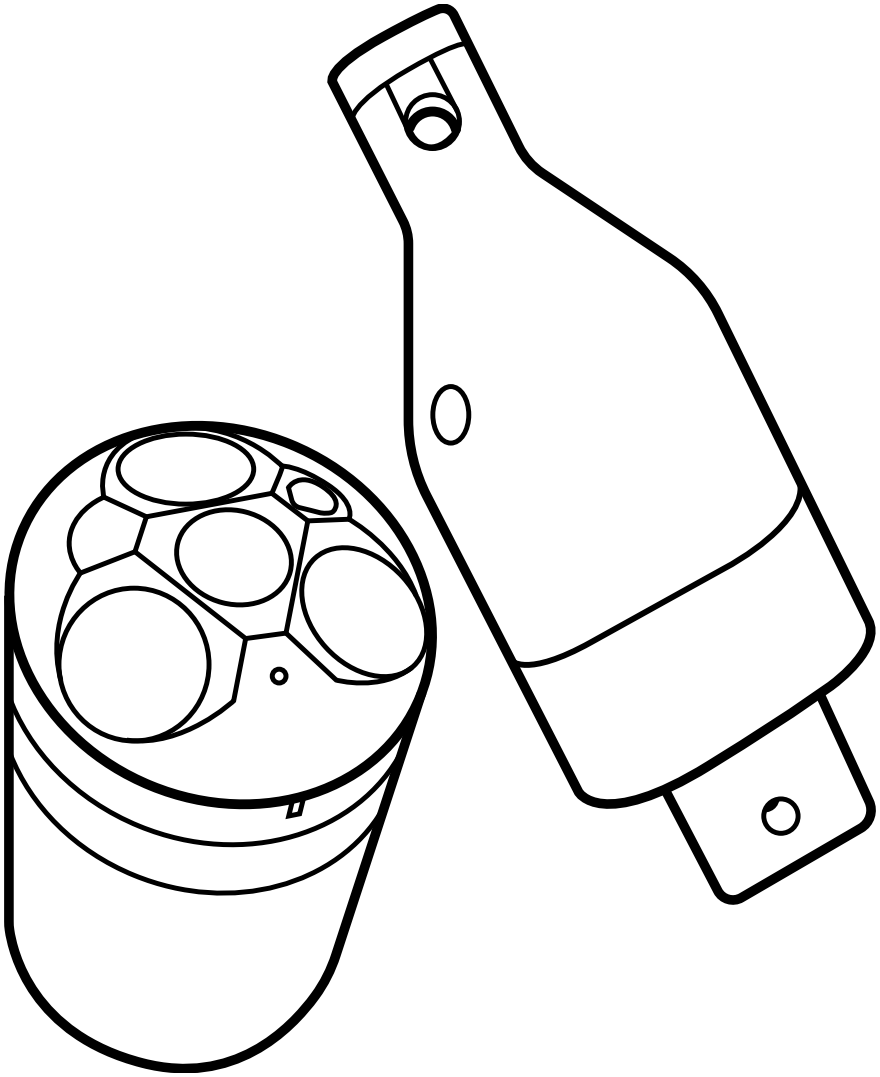
Eco Deployment Report



PSL
Renato Morais

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Mission

This section contains basic mission information, including information about the instrument used in this deployment mission, a location map and basic data statistics from the mission. All times are referenced to the local time of the device used to configure the Eco. All depths are referenced to the water surface

General info

Deployment/File Name: Eco773_202511180838			
Operator: Renato Morais			
Full Deployment Date Range: 18/11/2025, 8:38:44 am to: 1/12/2025, 2:38:44 pm			
ID	Eco_0773	Firmware	8008.8
Samples	9533	Interval (minutes)	2
Salinity	Salt water	35 ppt	
Latitude	-5.409491	Longitude	53.298512
Magnetic declination	-5.32419		
Average direction	70.5 deg	Average speed	0.17 m/s
Max Depth Below Surface	20.03 m		
Upper layer depth	9.28 m		
Middle layer depth	12.86 m		
Lower layer depth	16.44 m		
User id	017DD75E-7131-4FBB-8C2E-4DD5498E9D62		
Data id	078005E7-825F-418C-94FD-3FFD8301D438		
Created	1/12/2025, 2:47:11 pm		
Processor version	2.1.1.0		
Processed	1/12/2025, 2:47:29 pm		

Statistics (for Report Date Range only)

		Upper	Middle	Lower	Unit
Speed	**Mean**	0.18	0.18	0.16	m/s
	Max	0.65	0.62	0.60	m/s
	Min	0.00	0.00	0.00	m/s
	Std. dev	0.12	0.11	0.10	m/s

	Mean	Max	Min	Std. dev	Unit
Temperature	27.8	37.0	9.7	1.1	°C
Pressure	18.9	20.1	0.0	2.2	m
Tilt	9.0	37.0	0.1	0.8	deg

Location

Deployment location: Latitude -5.409491 and Longitude 53.298512.



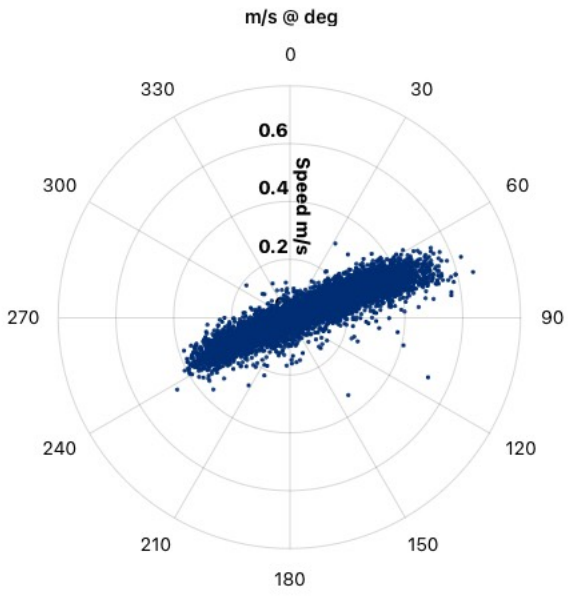
Tiles © Esri — Source: Esri, i-cubed, USDA, USGS, AEX, GeoEye, Getmapping, Aerogrid, IGN, IGP, UPR-EGP, and the GIS User Community

Figures

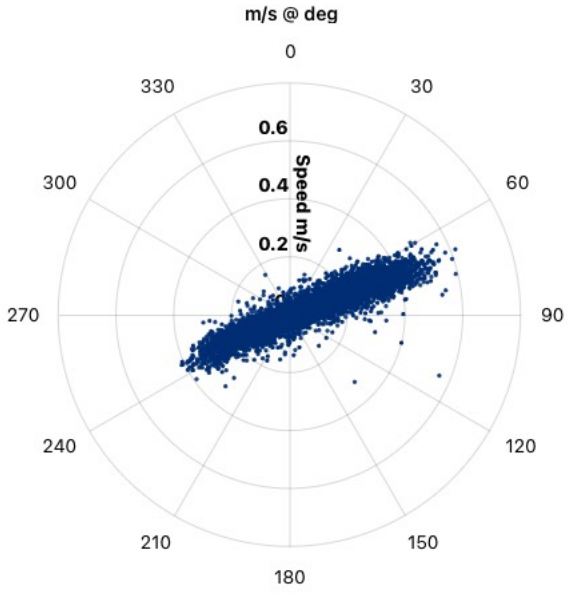
Polar Scatter Plots

These plots show the current speed and direction in a polar coordinate graph (North at the top, 0°) for each of the three depth layers. Data is shown for the Report Date Range. The numbers on the outside of the graph indicate the direction the current is flowing towards for that layer. For example, if a dot is plotted along the 40° line, then it indicates current flowing towards the Northeast. The distance away from the center of the graph indicates how fast the current is flowing: the closer to the center, the slower the water is moving.

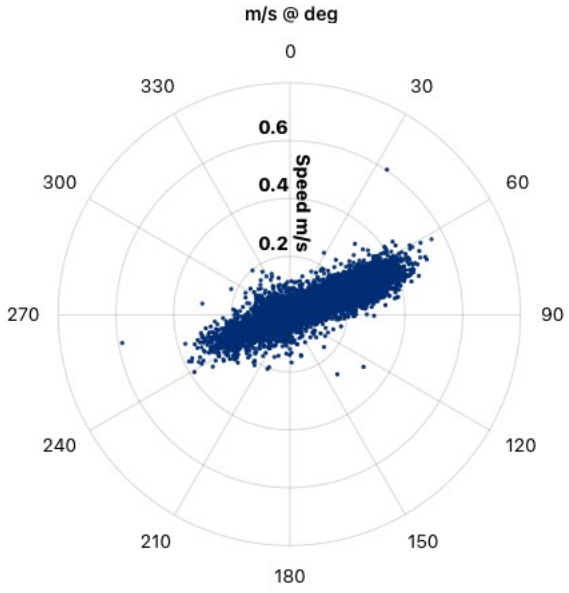
Upper Layer 9.28 m



Middle Layer 12.86 m



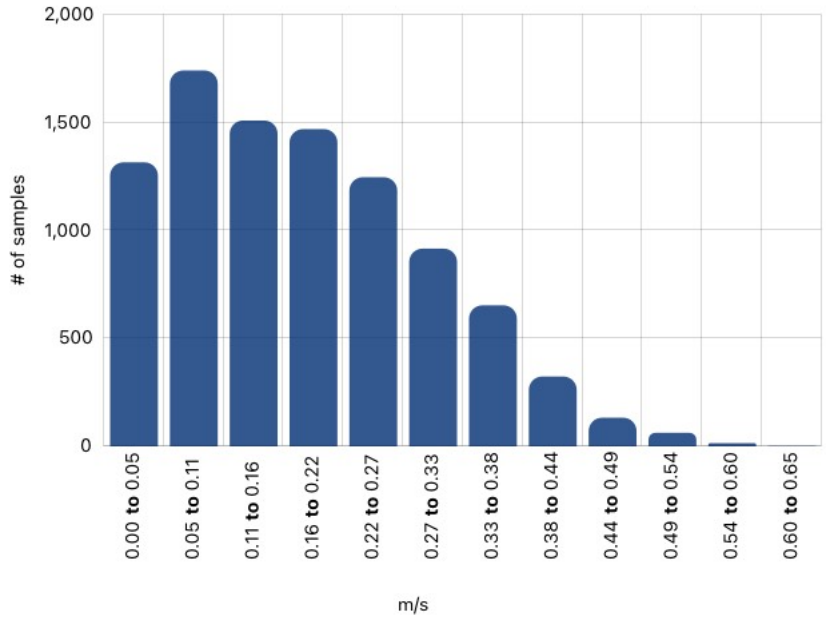
Lower Layer 16.44 m



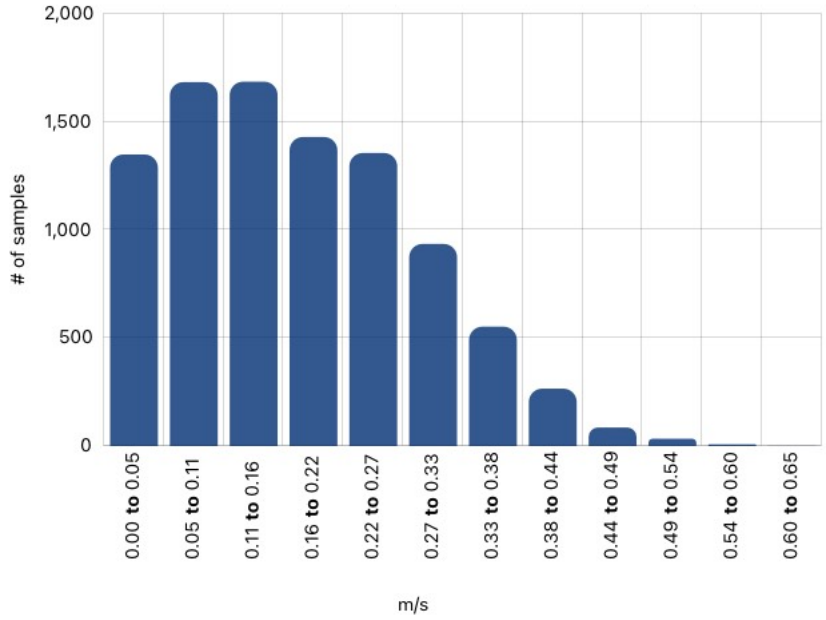
Histogram of Speed

These plots indicate the distribution of current speed for each of the three depth layers. The graph is generated by first taking the current speed data from the Report Date Range and dividing it into intervals of equal size. Then, the number of samples (i.e. an individual measurement interval) in each interval is counted. The numbers on the bottom (X-axis) indicate the speed intervals. The numbers on the left (Y-axis) indicate the number of samples in each interval. For example, if the highest bar on the graph is from the 0.3-0.4 m/s range, then that means that the current speed in this range happens more often than any other speed in the Report Date Range.

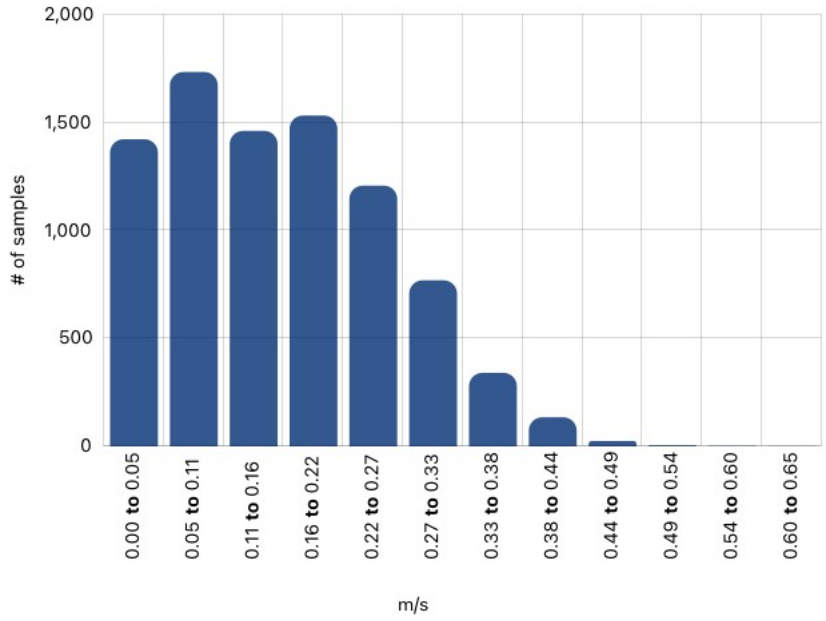
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Middle Layer 12.86 m



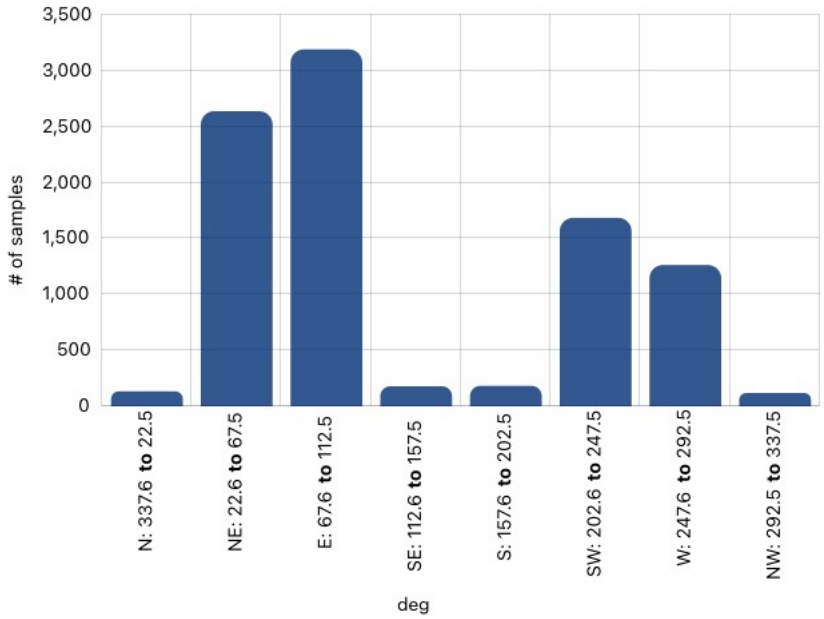
Lower Layer 16.44 m



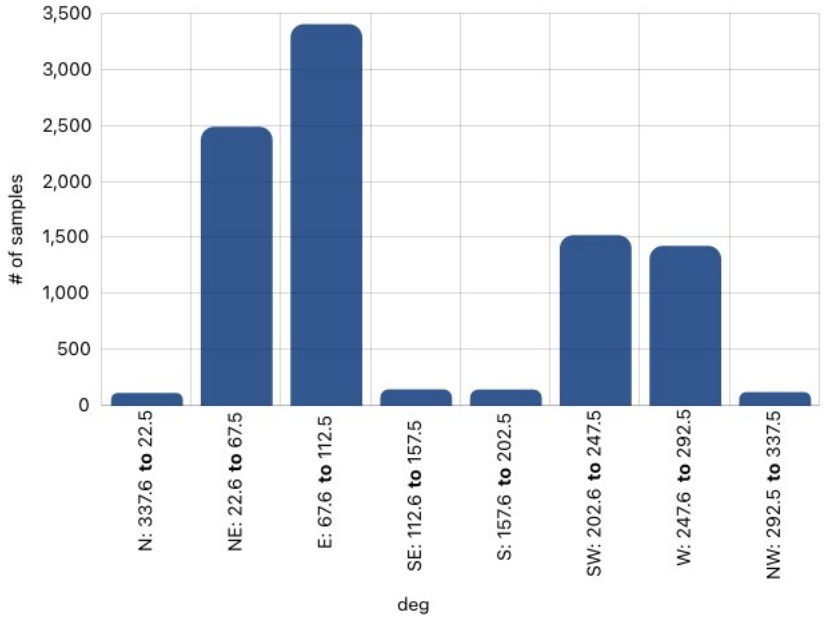
Histogram of Direction

These plots indicate the distribution of current direction for each of the three depth layers. The graph is generated by first taking the current direction data from the Report Date Range and dividing it into ranges of equal size. Then, the number of samples (i.e. an individual measurement interval) in each range is counted. The numbers on the bottom (X-axis) indicate the direction ranges. The numbers on the left (Y-axis) indicate the number of samples in each range. For example, if the highest bar on the graph is from the 90-120° range, then that means that the current direction in this range happens more often than any other direction during the Report Date Range.

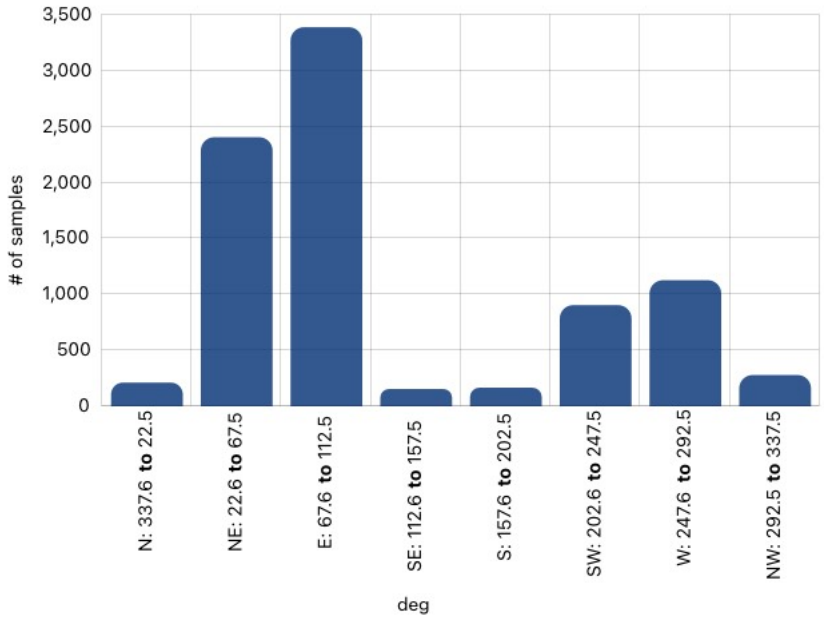
Upper Layer 9.28 m



Middle Layer 12.86 m



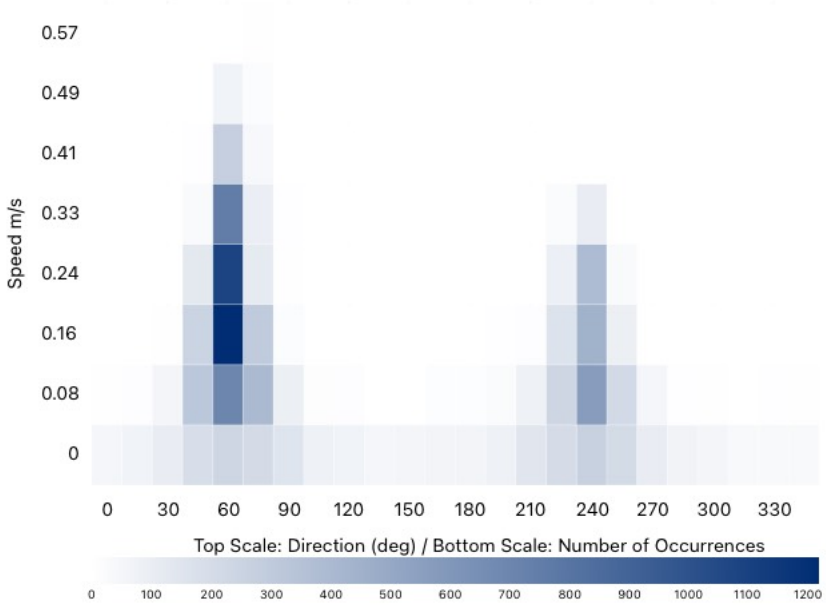
Lower Layer 16.44 m



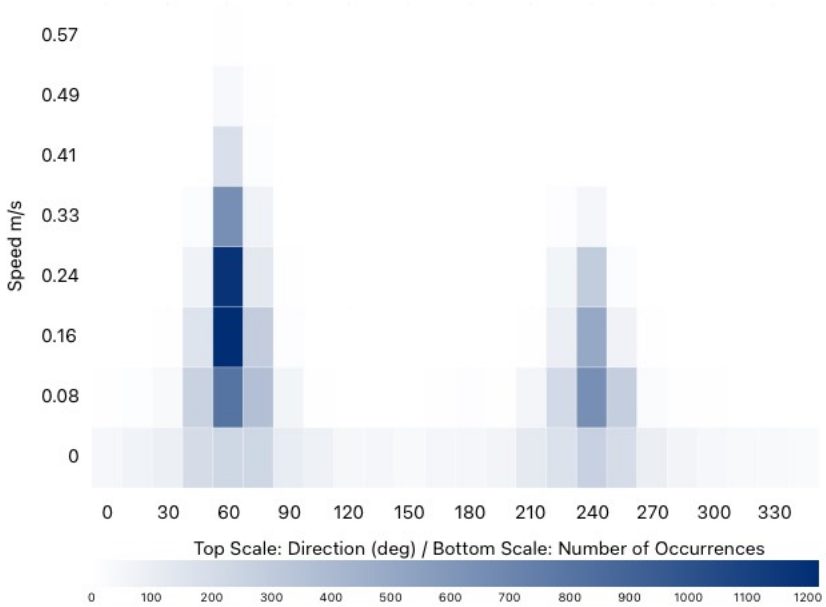
Speed/Direction Occurrence Frequency

These plots combine the speed and direction histograms to show the occurrence frequency of current speed and direction for each of the three depth layers in a cartesian coordinate graph. Data is shown for the Report Date Range. The numbers on the left (Y-axis) indicate the current's speed while the numbers along the lower (X-axis) indicate the direction the current is flowing towards. The color of each block inside the graph indicates the occurrence frequency of that particular speed/direction combination, scaled by the color bar on the bottom. For example, if the block with the darkest color is the intersection of 270° and 0.3 m/s, then that means that the currents are moving at this speed and direction more often than any other speed/direction.

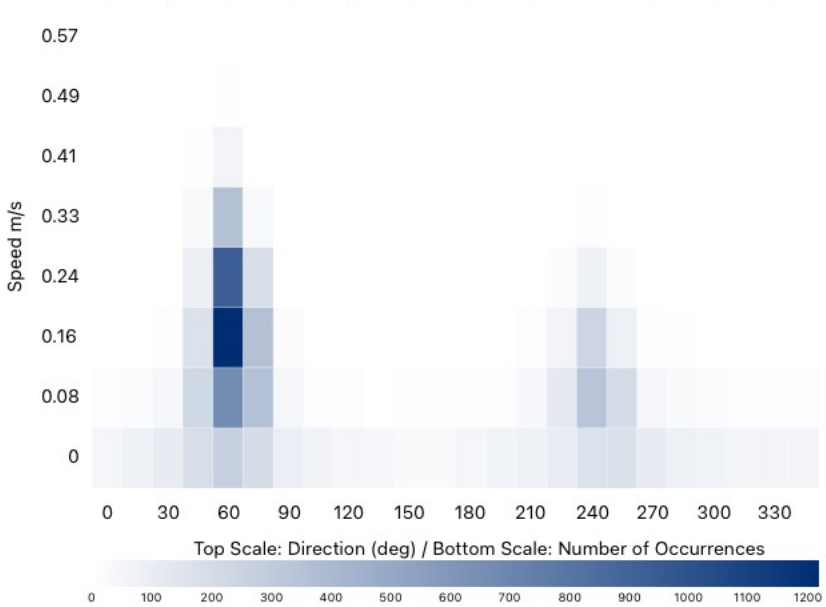
Upper Layer 9.28 m



Middle Layer 12.86 m



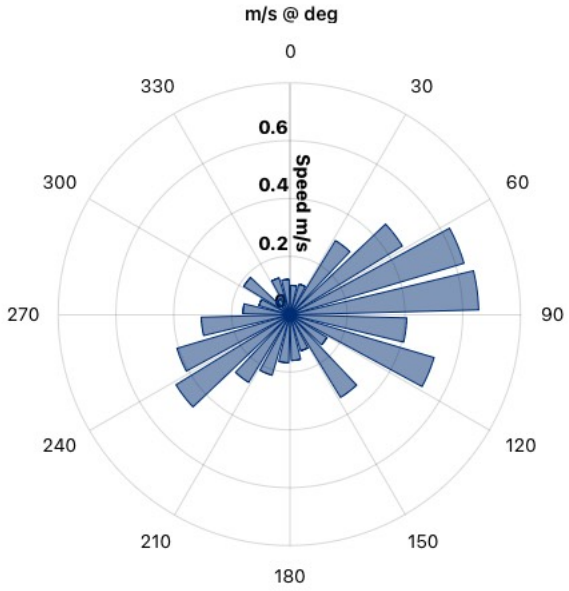
Lower Layer 16.44 m



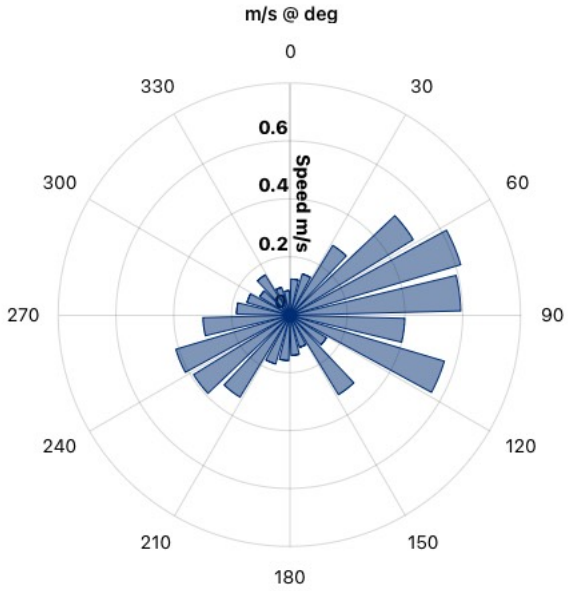
Direction and Maximum Speed

Similar to the Polar Scatter plots, these plots show the current speed and direction in a polar coordinate graph (North at the top; 0°) for each of the three depth layers. However, here only the maximum current speed, and its corresponding direction, are shown. The data are plotted as bars extending from the center of the graph. For example, if the longest bar (the one extending the farthest from the center of the graph) is at 270°, then that means the highest current speed was measured along this direction for this particular depth layer.

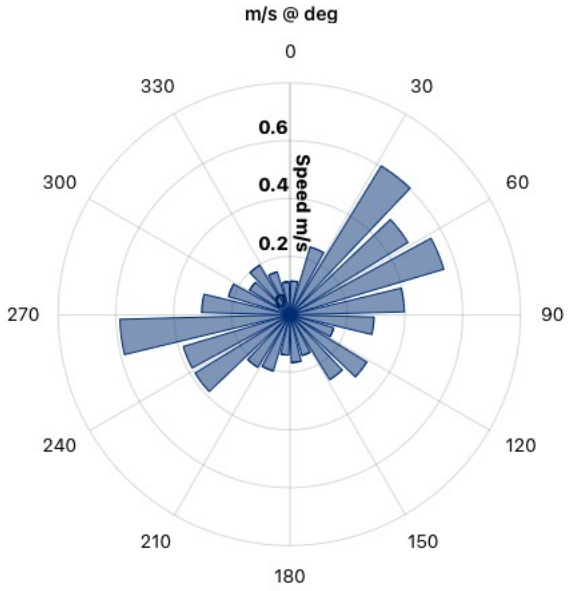
Upper Layer 9.28 m



Middle Layer 12.86 m



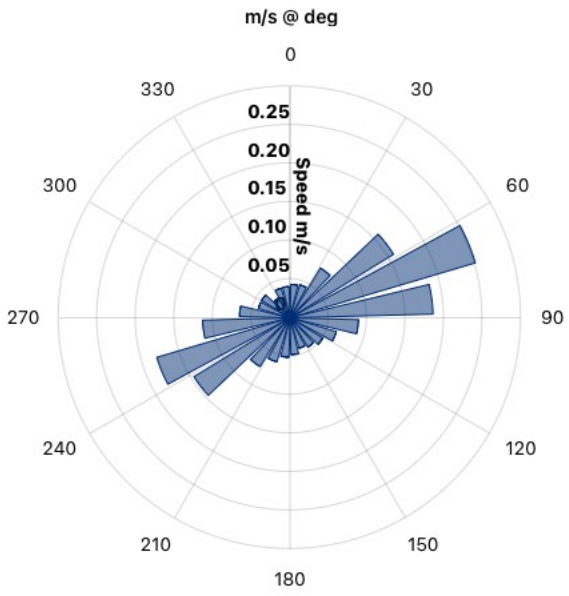
Lower Layer 16.44 m



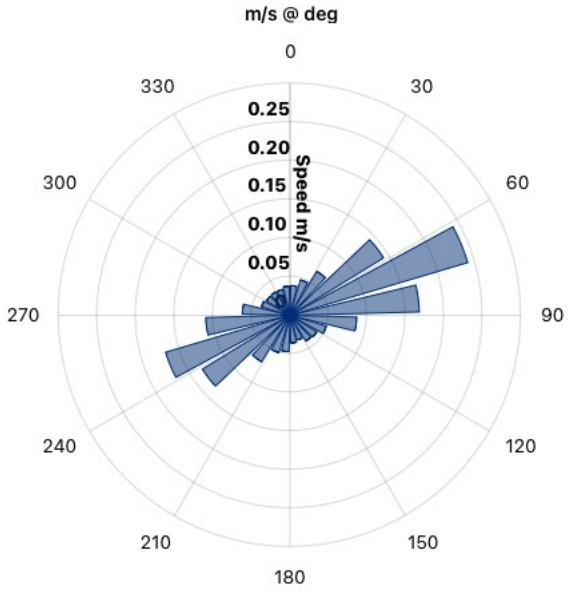
Direction and Mean Speed

Similar to the Polar Scatter plots, these plots show the current speed and direction in a polar coordinate graph (North at the top; 0°) for each of the three depth layers. However, here only the mean current speed, and its corresponding direction, are shown. The data are plotted as bars extending from the center of the graph. For example, if the longest bar (the one extending the farthest from the center of the graph) is at 180°, then that means the mean (average) current speed was measured along this direction for this particular depth layer.

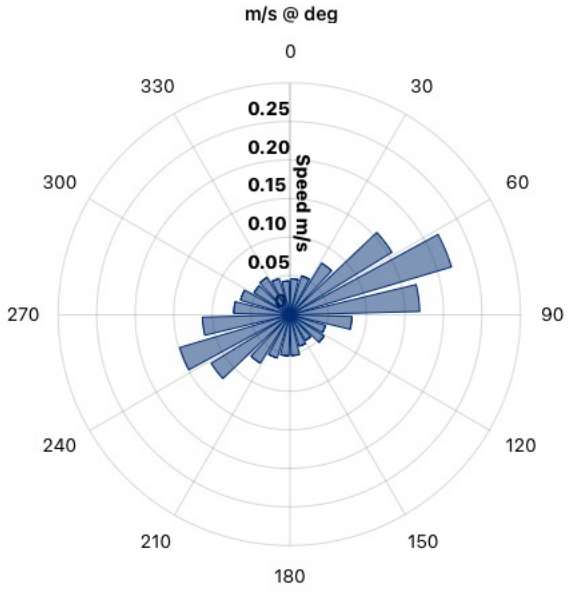
Upper Layer 9.28 m



Middle Layer 12.86 m



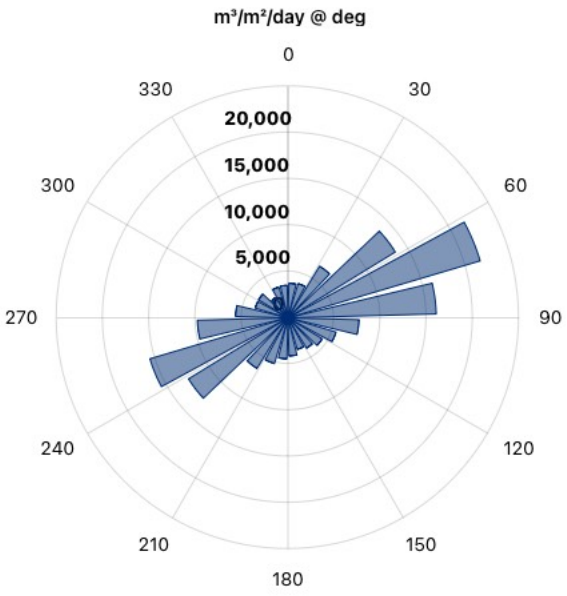
Lower Layer 16.44 m



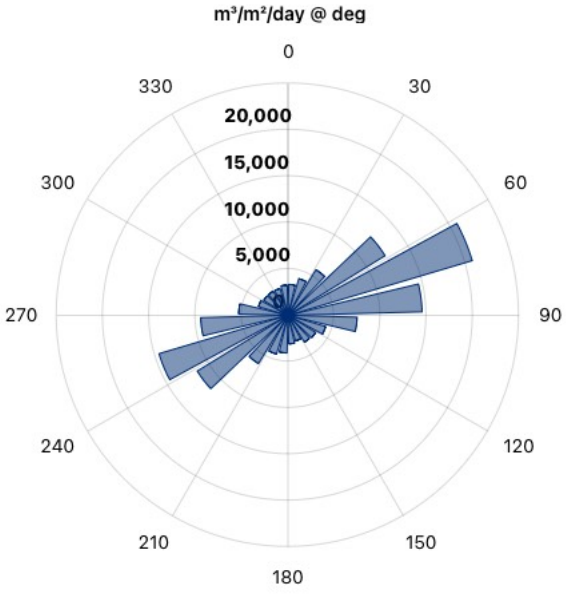
Direction and Flow

These plots show the average volume of water per day (flow) over the cross sectional area of the acoustic cone above the instrument. Data has units of $m^3/m^2/day$ (cubic meter per square meter per day). The data are plotted as bars extending from the center of the graph. For example, if the longest bar (the one extending the farthest from the center of the graph) is at 320°, and the distance from the center of the graph is along the 10,000 inner circle, that means that an average of 10,000 m3 of water per day flows across the acoustic cone of the instrument for that particular depth layer.

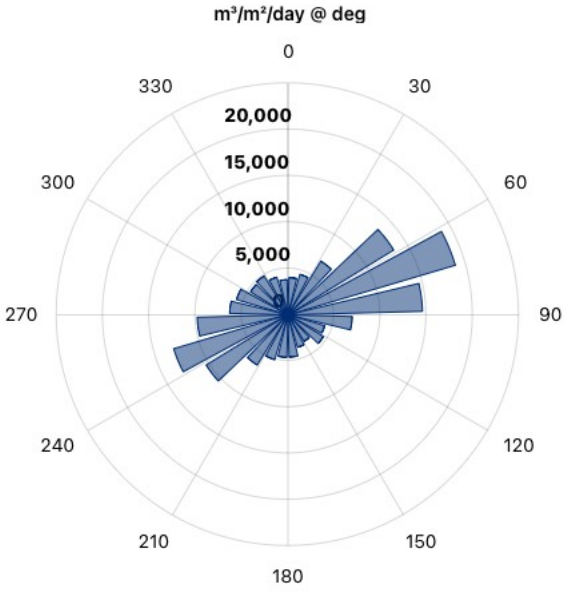
Upper Layer 9.28 m



Middle Layer 12.86 m



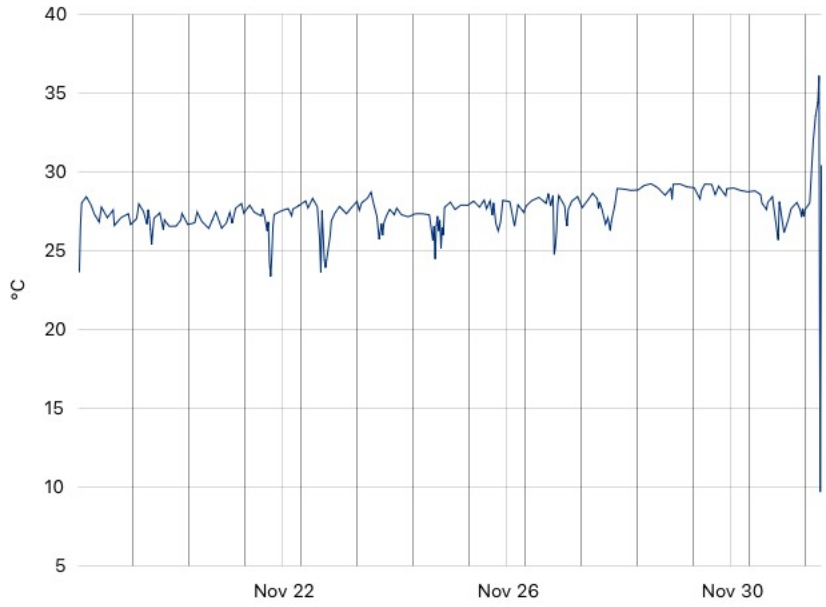
Lower Layer 16.44 m



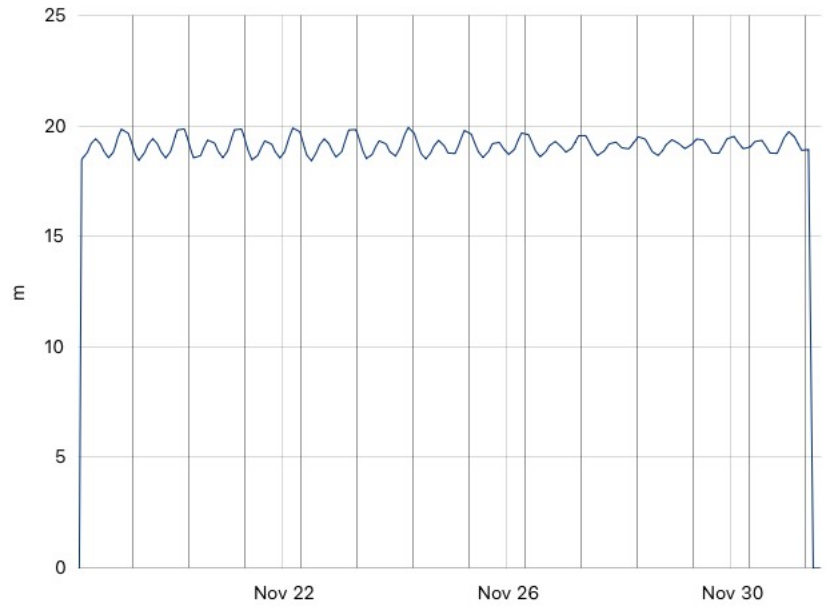
Environment

Three plots provide details about the environment over time (i.e. a time-series). These include: water temperature, water pressure (depth) and instrument tilt angle. Data is shown for the Report Date Range. The number on the bottom (X-axis) indicate the date/time while the numbers of the left (Y-axis) indicate the respective data value.

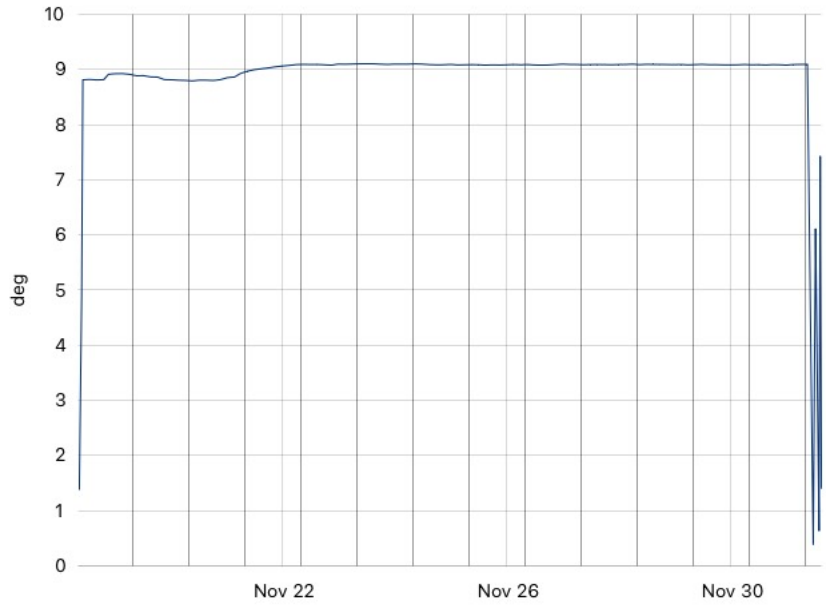
Temperature



Pressure



Tilt



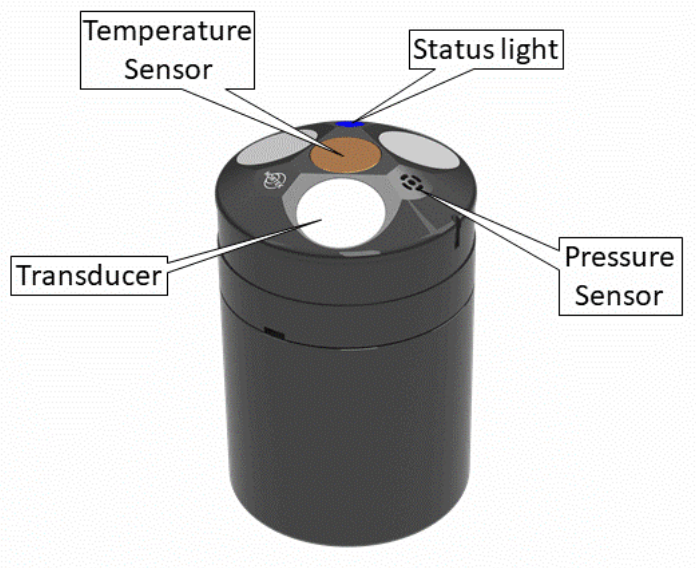
services-pdf-report.Deployment-Comments

Nortek Eco

Nortek designs, develops and produces scientific instruments that apply the Doppler principle to underwater acoustics in order to measure water in motion, such as currents and waves. We provide truly innovative, robust, and accurate instruments, backed up by advanced software and comprehensive support to ensure customers maximize value from their measurements. To learn more about Nortek, please visit our website at

www.nortekgroup.com.

The Nortek Eco is a class of oceanographic instruments called Acoustic Doppler Current Profilers (or ADCPs). As the name suggests, an ADCP uses the acoustic Doppler principle to measure current speed and direction at different depths, all from a single instrument. The Nortek Eco is the world's first portable ADCP and it was designed specifically to measure currents in shallow waters (up to 20 m depth).



The Nortek Eco and its main components.

Measurement principle

To measure water speed, the Nortek Eco transmits acoustic pulses that propagate upward through the water column. These pulses reflect off the particles in the water, generating echoes that are detected by the instrument. From this detection, the Eco uses the Doppler principle to compute the current's speed and direction. The acoustic pulses transmitted by the Eco originate from the three white circles (called transducers) on the instrument's top.

How to interpret the data

Data from the Nortek Eco can be read directly from the graphs generated in this report. All of the data have already been processed so no further processing is required. The Nortek Eco has a maximum profiling range of 20 m. This means that it can detect the current speed and direction as far as 20 m above it.

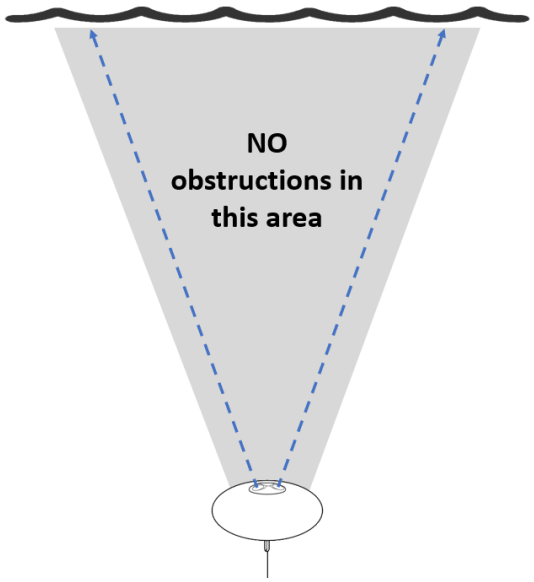
The water column above the Eco is divided into three layers: Upper, Middle and Lower. The Lower layer is the one closest to the Eco and the Upper layer is the one farthest from the Eco (that is, closest to the surface). Both the WebApp and this report provide the depth for each layer. For further details, see the next Section.

The velocity inside each layer is referenced to Earth coordinates, with direction being the direction that the current is flowing to. So for example, a velocity of 0.6 m/s at 90° for the upper layer means the top of the water column is flowing to the East at 0.6 m/s.

The Nortek Eco also records water pressure (depth), water temperature and its own tilt. These data are provided as time series along with the current speed data.

Sources of error

The Nortek Eco automatically processes all data collected, excluding data that do not meet quality standards, so you can be sure you are getting valid data for your deployment missions. Despite this robust internal processing, some conditions may increase the chances of collecting low quality data and should be avoided. These include: high tilt, blocked pressure sensor and blocked transducer. To minimize these effects, please ensure the following conditions are met for each Eco deployment mission:



Gray area (acoustic cone) must be free from obstructions during a mission. Angle of cone is 40°.

END OF REPORT